2 Lecture 2 (January 13, 2025): Affine groups and Hopf algebras

Scribe: Justin Bloom

More on affine group schemes

Last time: We defined affine schemes over k as *representable* functors

$$X : k\text{-}\mathsf{Alg} \to \mathsf{Set},$$

represented by a coordinate algebra denoted $k[X]$ (or $\mathcal{O}(X)$). The Yoneda lemma establishes a contravariant equivalence (or antiequivalence) of categories

$$\mathsf{AffSch}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} k\text{-}\mathsf{Alg}\ ,$$

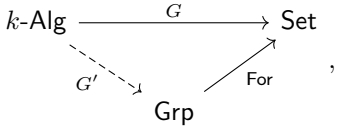
where the functor Spec is defined by $\operatorname{Spec} A : B \mapsto \operatorname{Hom}_{k\text{-}\mathsf{Alg}}(A, B)$.

Today: We will define the category of *affine group schemes* over k, denoted AffGrp_k, as well as the category of *Hopf algebras* over k, denoted Hopf_k. We will see that the antiequivalence above descends to a contravariant equivalence

$$\mathsf{AffGrp}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} \mathsf{CommHopf}_k\ ,$$

where $\mathsf{CommHopf}_k$ is the category of commutative Hopf algebras and Hopf-algebra homomorphisms. Its morphisms preserve multiplication, unit, comultiplication, counit and antipode.

2.1 Definition. An *affine group scheme* is an affine scheme G which factors through the category of groups, i.e. such that there exists a functor G' making the diagram commute



,

where For is the forgetful functor taking any group to its underlying set. A *homomorphism* of affine group schemes $G_1 \to G_2$ is a natural map of diagrams, i.e. a natural map such that each $G_1'(R) \to G_2'(R)$ is a group homomorphism for any algebra R.

We will adopt the convention that the forgetful functor is ignored, and an affine group scheme G has target category Grp for simplicity.

2.2 Example. Consider $\mathbf{pt} \in \mathsf{AffSch}_k$, represented by k as an algebra over itself. We have $\mathbf{pt}(R) = \operatorname{Hom}(k, R)$ consists of a single point $\{*\}$, the structure map $k \to R$ included in defining R as an algebra over k, for any R. Since $\{*\}$ can always be given the structure of the trivial group, making any map $\mathbf{pt}(R) \to \mathbf{pt}(R')$ a group homomorphism, we have $\mathbf{pt}$ is an affine group scheme we may call $\mathbf{1}$.

2.3 Definition. Given $X, Y \in \mathsf{AffSch}_k$, we define

$$\begin{aligned} X \times Y : k\text{-}\mathsf{Alg} &\to \mathsf{Set} \\ R &\mapsto X(R) \times Y(R), \end{aligned}$$

called the *fiber product* of X and Y over k.

Notes:

1) We have $X \times Y \in \mathsf{AffSch}_k$, with representing algebra $k[X] \otimes k[Y]$, because it is a fact that $\otimes$ is the coproduct for commutative algebras over k.

2) The product $\times$ is the direct product in AffSch_k, as can be seen by the antiequivalence. In fact, it agrees with the direct product in the big category of Set-valued functors, as can be checked fiberwise.

2.4 Lemma. *If $G \in \mathsf{AffGrp}_k$, then $k[G] \in \mathsf{Hopf}_k$ (the coordinate algebra is a Hopf algebra).*

Proof. Since we have a functor

$$G : k\text{-}\mathsf{Alg} \to \mathsf{Grp},$$

we get maps in AffSch_k

$$G \times G \xrightarrow{\text{mult}} G, \qquad \mathbf{pt} \xrightarrow{\text{unit}} G,$$

$$G \xrightarrow{\text{inverse}} G.$$

The antiequivalence now gives us homomorphisms of algebras

$$k[G] \otimes k[G] \xleftarrow{\Delta} k[G], \qquad k \xleftarrow{\varepsilon} k[G],$$

$$k[G] \xleftarrow{S} k[G].$$

The associative, unit, and inverse axioms all have diagrams, which upon reversing arrows, give the axioms for coassociative, counit, and antipodes, which define a Hopf algebra. ∎

2.5 Theorem. *The Yoneda lemma establishes an antiequivalence*

$$\mathsf{AffGrp}_k \underset{\text{Spec}}{\overset{k[\ ]}{\rightleftarrows}} \mathsf{CommHopf}_k \ .$$

Proof. An immediate corollary of Lemma 2.4 is that the functor $k[\ ]$ takes any group scheme to a commutative Hopf algebra. We must argue for any $A \in \mathsf{CommHopf}_k$, that $\operatorname{Spec} A$ is an affine group scheme and that the relevant compositions are naturally isomorphic to the identity functors.

Let $G = \operatorname{Spec} A$ for a Hopf algebra A, and let $R \in k\text{-}\mathsf{Alg}$.

1) Let $f, g \in G(R)$. We'll define the group product fg as the composition

$$fg : A \xrightarrow{\Delta} A \otimes A \xrightarrow{f \otimes g} R \otimes R \to R,$$

with the rightmost map being multiplication for R, which for commutative R is indeed a homomorphism of algebras. Hence $fg \in G(R)$.

2) The unit in $G(R)$ is $A \xrightarrow{\varepsilon} k \to R$.

3) The inverse of $f \in G(R)$ is defined by

$$f^{-1} : A \xrightarrow{S} A \xrightarrow{f} R.$$

Since A is commutative, S and hence $f \circ S$ are algebra maps. Coassociativity gives associativity of this product, and the counit identities give the unit. In Sweedler notation,

$$\begin{aligned}(ff^{-1})(a) &= \sum f(a_1) f(S(a_2)) \\ &= f\Big(\sum a_1 S(a_2)\Big) = \epsilon(a) 1_R.\end{aligned}$$

The other antipode identity similarly gives $f^{-1} f = \epsilon(-)1_R$. For an algebra map $h : R \to R'$, composition $f \mapsto h \circ f$ preserves this product and the identity. Thus G is a group-valued functor.

What's left to check is that the group structures and Hopf algebra structures are recovered from the constructions we've defined. In other words, given G, we have $A = k[G]$ is a Hopf algebra as constructed in Lemma 2.4, and must verify that our construction for a group scheme $\operatorname{Spec} A$ recovers G. Similarly, starting with A a commutative Hopf algebra, we have now that $G = \operatorname{Spec} A$ is a group scheme, so we must verify that the Hopf algebra construction of 2.4 recovers the given structure on A. We complete this verification in Lecture 3. ■

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.